

Code Break Cipher Relay

Crack ten ciphers and race the answer to the judge

At a glance: Crack ten ciphers and race the answer to the judge before time runs out. Plan about 60 minutes for the core block; 4 to 6 teams of 4 to 5 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: substitution ciphers, encodings, and pattern recognition (Understand).
- Choose the right decoding tool for an unknown code and apply it accurately (Apply).
- Split a decoding task across teammates and reassemble the parts into one message (Analyze).
- Verify a decoded message against its rule before racing it to the judge (Evaluate).

MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- Printed decoder sheets (six, one per team)
- Printed code slips (ten pages, each cut into six slips)
- Staff-only answer key (in the PY-STEM instructor answer key pack; never posted)
- Pencils and scratch paper
- Timer

PREP BEFORE STUDENTS ARRIVE (10 TO 15 MIN)

- 01 Print the decoder sheet six times (one per team) and the ten code-slip pages once, single-sided.
- 02 Cut each code page into its six slips and stack them face down by round.
- 03 Set up the judge table with the staff-only answer key and a score board; keep both out of student sight.
- 04 Clear a walking lane from the team tables to the judge so runners never cross a work area.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Crack ten ciphers and race the answer to the judge." Decode one two-letter Caesar example together on the board.

Phase 2 Decoder tour

0:05 - 0:10

Walk the decoder sheet: the four tables (numbers, Morse, grid, binary) and the four letter rules (shift, mirror, backwards, first letters). Teams write their runner rotation on the handout.

Phase 3 Rounds 1 to 5

0:10 - 0:32

Hand each team its slip face down, call GO, and time four minutes. Log the finish order as correct answers arrive; a wrong answer gets "not yet" and unlimited retries. Unsolved at time scores zero for that code.

Phase 4 Rounds 6 to 10

0:32 - 0:55

Same rhythm, harder codes. Remind teams to split the symbols across teammates; the final code hides two codes on one slip.

Phase 5 Score, debrief, cleanup

last 5 min

Total each team's points (a perfect sweep is exactly 100), ask one debrief question, and collect the slips and decoder sheets.

TROUBLESHOOTING

Problem: Two runners reach the judge in a dead heat. **Fix:** Both teams take the higher finish slot and the next slot is skipped; log it and move on.

Problem: A runner says the answer out loud and other teams hear it. **Fix:** Enforce whisper-or-show-it-written at the judge table; keep the finish order as logged and remind runners quietly.

Problem: A team is stuck at zero and deflating. **Fix:** Point them at the hint printed on the slip and the matching decoder table; the tier prompts below un-stick a team without giving the answer away.

Problem: A team finishes early every round. **Fix:** Push the extension prompt below and have them write their own coded message for a judge to crack.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "Which table on the decoder sheet matches the symbols on your slip?" Point them to the hint line on the slip.

Tier 2 (decoding but slowly):

- "Split the message: who takes the first half of the symbols, who takes the second?"

Tier 3 (ready to extend):

- "Write your own five-word message in the hardest code you beat and trade it with another finished team."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which code fell fastest once you knew the rule, and why?
- How did splitting the symbols across teammates change your speed?
- Where does the real world still use codes like these?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
First correct team on a code	10
Second to fifth correct team	8, 6, 4, 2
Any later correct answer before time	1
Unsolved when time is called	0
Total · perfect sweep of ten codes	100

CLEANUP AND SOURCES

Collect the code slips and decoder sheets for the next group, return pencils, and log each team's total on the score board.

SOURCES Classic classroom ciphers (Caesar, Atbash, Polybius, Morse); computer science unplugged decoding relays